

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - V

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Class : SYBSCIT	Batch : 1
Date of Assignment : 22-07-2026	Date/Time of Submission : 22-07-2026

Binary Search Tree: Write a program to:

- a. Create a binary search tree.

Code:

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
nodeA = TreeNode('A')
nodeB = TreeNode('B')
nodeC = TreeNode('C')
nodeD = TreeNode('D')
nodeE = TreeNode('E')
nodeF = TreeNode('F')
nodeG = TreeNode('G')

root.left = nodeA
root.right = nodeB

nodeA.left = nodeC
nodeA.right = nodeD

nodeB.left = nodeE
nodeB.right = nodeF

nodeF.left = nodeG

print("root.right.left.data:", root.right.left.data)
```

Output:

```
= RESTART: C:/Users/nidhi/Downloads/Nidhi DS/tree.  
py  
root.right.left.data: E
```

b. Insert nodes into a binary search tree.

Code:

```
nodeH = TreeNode('H')  
  
nodeF.left = nodeG  
nodeF.right = nodeH  
  
# Test  
print("", root.right.right.right.data)
```

Output:

```
= RESTART: C:/Users/nidhi/Downloads/Nidhi DS/tree.  
py  
H
```

c. Search for a node in a binary search tree.

Code:

```

def search(root, key):
    if root is None:
        return False

    print("Checking: ", root.data)

    if root.data == key:
        return True

    return search(root.left, key) or
search(root.right, key)

key = input("Enter Element to search: ")

if search(root, key):
    print("Element Found")
else:
    print("Element Not Found")

```

Output:

```

= RESTART: C:/Users/nidhi/Downloa
py
H
Enter Element to search: H
Checking:  R
Checking:  A
Checking:  C
Checking:  D
Checking:  B
Checking:  E
Checking:  F
Checking:  G
Checking:  H
Element Found
|

```